

# M2S-99: Best Practice Summary

**Series:** M2S — Managed to SaaS Migration | **Notebook:** 99 | **Created:** March 2026 | **Last Updated:** 07/24/2026

A definitive, actionable reference of every best practice extracted from the M2S Managed-to-SaaS Migration series (notebooks 01–09). Each practice specifies the exact setting or action, its priority, and its step. No hedging—follow these and your migration succeeds.

## The 9-Step Migration Framework

The M2S series follows a structured, step-based approach to Managed-to-SaaS migration:

| Step | Name       | Focus                                |
|------|------------|--------------------------------------|
| 1    | Discover   | Understand SaaS Differences          |
| 2    | Strategize | Define Your Migration Approach       |
| 3    | Design     | Create Target Architecture           |
| 4    | Prepare    | Readiness and Pre-Migration          |
| 5    | Execute    | Migrate Configuration and Agents     |
| 6    | Integrate  | Reconnect Integrations               |
| 7    | Enable     | User Enablement and Communication    |
| 8    | Expand     | Adopt New SaaS Capabilities          |
| 9    | Optimize   | Validate, Optimize, and Decommission |

**OneAgent Attribute Enrichment (1.333+):** OneAgent can enrich all telemetry (metrics, spans, logs, events) with primary fields ( `dt.security_context` , `dt.cost.costcenter` ) and primary tags ( `primary_tags.environment` , `primary_tags.team` ) at the source. More efficient than auto-tags — feeds directly into OpenPipeline routing, bucket assignment, and Grail permissions. Configure via `oneagentctl --set-host-tag` or `--set-host-tag` at install time. See [docs](#).

## The 11-Step Order of Operations (Upgrade Phase)

Within the upgrade phase, follow this precise execution order:

| #  | Operation           | Description                                      |
|----|---------------------|--------------------------------------------------|
| 1  | <b>Assess</b>       | Inventory current environment                    |
| 2  | <b>Provision</b>    | SaaS tenant and access (SSO)                     |
| 3  | <b>Install</b>      | New ActiveGates in parallel with old ActiveGates |
| 4  | <b>Migrate</b>      | Configuration and integrations                   |
| 5  | <b>Rebuild</b>      | Dashboards, zones, alerts                        |
| 6  | <b>Redirect</b>     | OneAgents to SaaS                                |
| 7  | <b>Reconnect</b>    | Integrations and extensions                      |
| 8  | <b>Migrate</b>      | Any remaining configuration and integrations     |
| 9  | <b>Validate</b>     | Data flow and performance                        |
| 10 | <b>Cutover</b>      | Full switch to SaaS                              |
| 11 | <b>Decommission</b> | Managed environment                              |

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## 1. Step 1: Discover — Understand SaaS Differences

| Practice                                  | Recommended Setting/Value                                                                                                                                                                                                   | Priority |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Complete entity discovery before strategy | Run Entities API v2 inventory: <code>type(HOST)</code> , <code>type(SERVICE)</code> , <code>type(APPLICATION)</code> , <code>type(SYNTHETIC_TEST)</code> . Document counts in a spreadsheet before any other planning work. | Critical |

| Practice                                       | Recommended Setting/Value                                                                                                                                                                                                                                                                                                                             | Priority    |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Plan for historic data loss                    | Historic metrics, logs, traces, problems, and session data <b>cannot be migrated</b> . Export critical reports before shutdown. Run dual environments to build SaaS baselines.                                                                                                                                                                        | Critical    |
| Plan tenant sizing against the host guideline  | Dynatrace recommends roughly <b>25,000 hosts per environment</b> (about 50,000 per cluster) as a sizing guideline — verify current thresholds with Dynatrace. The enforced ceiling is your licensed <b>host-unit quota</b> , a separate metric. Split into multiple environments when consolidating Managed clusters would exceed the recommendation. | Recommended |
| Align Managed and SaaS versions                | There is <b>no fixed minimum Managed version</b> for the SaaS Upgrade Assistant — align the Managed cluster to the <b>same major version</b> as the target SaaS tenant rather than targeting a specific floor.                                                                                                                                        | Critical    |
| Use SaaS Upgrade Assistant for most migrations | Install the app on the target SaaS tenant. Assign <code>upgrade-assistant:environments:write</code> IAM policy to migration users. The SaaS Upgrade Assistant is the recommended tool for the majority of migrations.                                                                                                                                 | Recommended |
| Use Monaco for tenant consolidation            | When consolidating multiple Managed tenants with <b>identical configuration names</b> , use Monaco—the SaaS Upgrade Assistant cannot resolve name conflicts.                                                                                                                                                                                          | Recommended |

## 2. Step 2: Strategize — Define Your Migration Approach

| Practice                                          | Recommended Setting/Value                                                                                                                                                                                                                                                                                                                                                          | Priority    |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Choose migration approach by environment size     | <500 hosts: Big Bang. 500–2,000 hosts: Phased by environment. >2,000 hosts or complex integrations: Phased by region/application.                                                                                                                                                                                                                                                  | Critical    |
| Define measurable success criteria                | Set targets: 100% host coverage, 100% service discovery, <15 min data gaps, 100% alert delivery, 100% dashboard availability, 100% integration success.                                                                                                                                                                                                                            | Critical    |
| Budget time for the 10% manual items (90/10 rule) | 90% of configs migrate automatically; the remaining 10% (credentials, webhooks, custom scripts) takes 90% of the manual effort. Plan accordingly.                                                                                                                                                                                                                                  | Recommended |
| Engage Dynatrace Professional Services early      | Involve account team and PS during the Strategize phase, not during execution. Early engagement prevents avoidable rework.                                                                                                                                                                                                                                                         | Recommended |
| Choose one migration tool                         | Pick <b>one</b> primary tool: SaaS Upgrade Assistant (recommended), Monaco, Terraform, or Settings API. Never mix approaches—Monaco YAML conflicts with the SaaS Upgrade Assistant. The rule is one writer per configuration schema per migration window; sequencing tools with an explicit ownership handoff is safe. For the Terraform path and the handoff pattern, see M2S-95. | Critical    |

### 3. Step 3: Design — Create Target Architecture

| Practice                                        | Recommended Setting/Value                                                                                                                                                                  | Priority    |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Open outbound 443 to SaaS endpoints             | Allow HTTPS (port <b>443</b> ) from all monitored hosts and ActiveGates to <code>{tenant-id}.live.dynatrace.com</code> and <code>{tenant-id}.apps.dynatrace.com</code> .                   | Critical    |
| Use ActiveGate routing for restricted networks  | If hosts cannot reach the internet directly, deploy Environment ActiveGates. OneAgents connect to AG on port <b>9999</b> ; AG connects outbound on <b>443</b> .                            | Critical    |
| Recreate Network Zones in SaaS before migrating | Network Zone configuration does <b>not</b> transfer automatically. Create zones in SaaS via Settings > Network zones or the API, then assign AGs and OneAgents.                            | Critical    |
| Deploy minimum 2 ActiveGates per network zone   | Provides high availability within each zone. OneAgent auto-discovers available AGs.                                                                                                        | Recommended |
| Configure SAML SSO with full message signing    | IdP must sign the <b>entire SAML message</b> , not just the assertion. Azure AD meets this requirement by default. Failure causes authentication errors.                                   | Critical    |
| Confirm data residency region at provisioning   | Select the correct region (US, EU, APAC) during tenant provisioning. <b>Cannot be changed</b> after provisioning.                                                                          | Critical    |
| Enforce TLS 1.2+ for all traffic                | Dynatrace SaaS enforces TLS 1.2 minimum. Verify no legacy TLS 1.0/1.1 configurations on proxies or load balancers.                                                                         | Critical    |
| Ensure DNS resolves SaaS domains                | Verify DNS resolution for <code>{tenant-id}.live.dynatrace.com</code> , <code>{tenant-id}.apps.dynatrace.com</code> , and <code>*.dynatrace.com</code> .                                   | Critical    |
| Match network zones to physical topology        | Name zones by datacenter or region (e.g., <code>datacenter-east</code> , <code>aws-us-east-1</code> ). Define alternative zones for failover.                                              | Recommended |
| Size ActiveGates by host count                  | Up to 500 hosts: <b>2 CPU / 4 GB RAM / 20 GB disk</b> . 500–1,500: <b>4 CPU / 8 GB / 40 GB</b> . 1,500–5,000: <b>8 CPU / 16 GB / 80 GB</b> . >5,000: scale horizontally with multiple AGs. | Recommended |

### 4. Step 4: Prepare — Readiness and Pre-Migration

| Practice                                   | Recommended Setting/Value                                                                                                                                                                                                                             | Priority |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Align the Managed version before upgrading | The SaaS Upgrade Assistant has <b>no fixed minimum Managed version</b> — align the Managed cluster to the <b>same major version</b> as the target SaaS tenant rather than a specific floor. Upgrade the Managed cluster first if the versions differ. | Critical |
| Coordinate dual licensing                  | Work with Dynatrace account team to arrange temporary overlap licensing for the period both Managed and SaaS run simultaneously.                                                                                                                      | Critical |

| Practice                                           | Recommended Setting/Value                                                                                                                                                               | Priority    |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Deploy new SaaS ActiveGates in parallel            | Install new SaaS-connected AGs alongside existing Managed AGs <b>before</b> migrating OneAgents. Validates connectivity without impacting monitored hosts.                              | Critical    |
| Install SaaS Upgrade Assistant                     | Install the app on the target SaaS tenant and verify connectivity to the Managed cluster. Confirm all required permissions are in place.                                                | Critical    |
| Announce configuration freeze                      | Notify all teams that no configuration changes should be made to the Managed environment during the migration window. Document the freeze start and end dates.                          | Recommended |
| Test rollback procedure                            | Document the Managed server URL and token. Rollback: <code>oneagentctl --set-server="https://{managed-cluster}/communication"</code> + restart. Test on one host before bulk migration. | Critical    |
| Test connectivity before migration                 | Run <code>curl -v https://{tenant-id}.live.dynatrace.com/api/v1/time</code> from monitored hosts and ActiveGate servers before scheduling the migration window.                         | Critical    |
| Verify OneAgent versions are within support window | OneAgent support window is <b>9 months (Standard) / 12 months (Enterprise)</b> . Check oldest deployed versions before migration. Upgrade outdated agents first.                        | Recommended |

## 5. Step 5: Execute — Migrate Configuration and Agents

| Practice                                                                 | Recommended Setting/Value                                                                                                                                                                                                                         | Priority |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Migrate configurations BEFORE redirecting OneAgents                      | Deploy foundational settings (management zones, tags, service detection rules) to SaaS before any agents start reporting.                                                                                                                         | Critical |
| Follow the 4-wave deployment order                                       | <b>Wave 1:</b> Management zones, auto-tagging rules, host groups. <b>Wave 2:</b> Service detection, request attributes, deep monitoring. <b>Wave 3:</b> Alerting profiles, anomaly detection, SLOs. <b>Wave 4:</b> Dashboards, remaining configs. | Critical |
| Use <code>oneagentctl --set-server</code> to reconfigure (not reinstall) | Run: <code>oneagentctl --set-server="https://{tenant}.live.dynatrace.com:443/communication"</code> then <code>--set-tenant-token="{token}"</code> then <code>systemctl restart oneagent</code> . Preserves host identity and custom metadata.     | Critical |
| Restart application processes after migration                            | Full-stack monitoring requires process restarts—Java, .NET, Node.js, PHP all inject instrumentation at process startup. Without restart: host metrics flow but <b>no distributed tracing, no service detection, no code-level visibility</b> .    | Critical |
| Migrate non-production first, then production                            | Validate data flow on non-prod hosts before touching production. Apply lessons learned to each subsequent wave.                                                                                                                                   | Critical |

| Practice                                        | Recommended Setting/Value                                                                                                                                                    | Priority    |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Download deploy result CSVs for audit           | After each wave, download the deployment result CSV from SaaS Upgrade Assistant. Archive these for compliance and troubleshooting.                                           | Recommended |
| Use SaaS Upgrade Assistant selective import     | Deploy configuration types incrementally using Smart Selective Import. Smart dependency management auto-adds required configs.                                               | Recommended |
| Preview all changes before deploying            | Use the SaaS Upgrade Assistant <b>Preview Changes</b> feature to verify edits. The app retains original values for rollback reference.                                       | Recommended |
| Manually recreate Credentials Vault entries     | Cannot be exported. Re-enter all credentials (cloud platform integrations, extension credentials) directly in SaaS.                                                          | Critical    |
| Generate new API tokens in SaaS                 | Managed tokens do not work in SaaS. Create new tokens with <b>minimal required scopes</b> for each use case.                                                                 | Critical    |
| Use Ansible/automation for bulk reconfiguration | For large environments, use configuration management (Ansible, Puppet, Chef) to execute <code>oneagentctl</code> commands and rolling application restarts across all hosts. | Recommended |

## 6. Step 6: Integrate — Reconnect Integrations

| Practice                                    | Recommended Setting/Value                                                                                                                                    | Priority    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Recreate webhooks with SaaS URLs            | Endpoint URLs change for SaaS. Create new Slack, Teams, PagerDuty, ServiceNow, and custom webhook integrations with SaaS URLs.                               | Critical    |
| Update all API base URLs                    | Change from <code>{managed-url}/e/{env-id}</code> to <code>{tenant}.live.dynatrace.com</code> . Update tokens in all CI/CD pipelines and automation scripts. | Critical    |
| Test alert delivery end-to-end              | Trigger a test alert and confirm delivery through every notification channel (email, Slack, Teams, PagerDuty, ServiceNow).                                   | Critical    |
| Recreate synthetic private locations        | Deploy new Synthetic ActiveGates in SaaS. Private locations are infrastructure-specific and do not transfer.                                                 | Recommended |
| Update CI/CD pipelines                      | Replace Managed API endpoints and tokens in all deployment pipelines, quality gates, and automated testing integrations.                                     | Critical    |
| Deploy dedicated AGs for Extensions 2.0     | Extensions 2.0 require <b>host-based</b> ActiveGate (not K8s-based). Deploy a separate AG group for extension execution.                                     | Recommended |
| Remove hardcoded entity IDs from dashboards | Before importing dashboards, replace hardcoded entity IDs with entity selectors. Update management zone IDs and dashboard owners.                            | Recommended |

## 7. Step 7: Enable — User Enablement and Communication

| Practice                                          | Recommended Setting/Value                                                                                                                                       | Priority    |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Communicate migration to all users                | Send advance notice with migration timeline, expected impacts, new login URLs, and support channels. Communicate before, during, and after each migration wave. | Critical    |
| Deliver persona-based training                    | Tailor training by role: SREs get DQL and alerting, developers get distributed tracing and code-level visibility, managers get dashboards and reporting.        | Recommended |
| Update all documentation                          | Replace Managed URLs, procedures, and screenshots with SaaS equivalents in runbooks, architecture diagrams, onboarding guides, and disaster recovery plans.     | Critical    |
| Establish support channels                        | Create dedicated Slack/Teams channels for migration questions. Designate migration champions in each application team.                                          | Recommended |
| Map Managed roles to SaaS IAM policies            | Cluster admin > Account admin. Environment admin > Environment admin. Monitor user > Viewer + specific policies. Custom roles > Custom IAM policies.            | Critical    |
| Replace LDAP with SAML                            | SaaS does not support direct LDAP authentication. Migrate to SAML 2.0 SSO through your IdP.                                                                     | Critical    |
| Filter SAML group claims to Dynatrace groups only | Azure Entra limit: <b>150 groups</b> per user in SAML claim. Filter to Dynatrace-related groups to stay under the limit.                                        | Recommended |

## 8. Step 8: Expand — Adopt New SaaS Capabilities

| Practice                                        | Recommended Setting/Value                                                                                                                                                                                           | Priority    |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Adopt Grail for unified querying                | Replace ad-hoc USQL queries with Notebook-based DQL analysis. Grail provides a unified data lakehouse for logs, metrics, traces, events, and entities.                                                              | Recommended |
| Configure OpenPipeline for log masking          | Configure log processing rules in OpenPipeline to redact PII (SSNs, credit card numbers, email addresses) before storage.                                                                                           | Recommended |
| Enable PII masking at ingest                    | Settings > Request attributes: enable masking on all attributes capturing user input, account numbers, or personal data. Configure session replay masking for RUM.                                                  | Critical    |
| Implement Workflows for automation              | Replace Managed problem notifications with SaaS Workflow triggers + HTTP Request actions. Recreate custom webhook logic as workflow steps.                                                                          | Recommended |
| Right-size data retention for cost optimization | Configure Grail bucket retention by data type. Logs: set retention based on compliance needs. Metrics: leverage the 5-year aggregated retention. Use <code>bucket:</code> targeting in queries to limit scan scope. | Recommended |

| Practice                                          | Recommended Setting/Value                                                      | Priority |
|---------------------------------------------------|--------------------------------------------------------------------------------|----------|
| Explore Dynatrace Assist for AI-assisted analysis | Enable Dynatrace Assist for natural language querying and root cause analysis. | Optional |

## 9. Step 9: Optimize — Validate, Optimize, and Decommission

| Practice                                             | Recommended Setting/Value                                                                                                                                                                                                      | Priority    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Allow 7–14 days for Dynatrace Intelligence baselines | Response time and error rate baselines: <b>2–7 days</b> . Resource usage and traffic patterns: <b>7–14 days</b> . Expect higher alert volume during this period—do not disable alerting.                                       | Critical    |
| Tune alert thresholds after baseline period          | After 2 weeks: raise thresholds on noisy alerts, add time-based conditions, lower thresholds on missing alerts.                                                                                                                | Recommended |
| Compare entity counts against inventory              | Run <code>fetch dt.entity.host \   summarize count()</code> (and <code>service</code> , <code>application</code> , <code>process_group</code> , <code>synthetic_test</code> ). Counts must match the planning-phase inventory. | Critical    |
| Verify metrics flow with no gaps >15 minutes         | Run <code>timeseries avg(dt.host.cpu.usage), from:-1h, by: {dt.entity.host}</code> and check for nulls. Any host returning null has a data gap.                                                                                | Critical    |
| Validate log ingestion is continuous                 | Run <code>fetch logs, from:-1h \   summarize count(), by: {bin(timestamp, 5m)}</code> and confirm no 5-minute buckets with zero count.                                                                                         | Critical    |
| Confirm distributed traces are flowing               | Run <code>fetch spans, from:-1h \   summarize count(), by: {bin(timestamp, 5m)}</code> . Zero spans after agent migration means application processes need restart.                                                            | Critical    |
| Get stakeholder sign-off                             | Obtain written approval from: Platform Team Lead, Security Team, Application Teams, Executive Sponsor.                                                                                                                         | Critical    |
| Keep Managed running 2–4 weeks post-migration        | Maintain access for historical reference and comparison. Decommission only after full validation period.                                                                                                                       | Recommended |
| Archive migration artifacts before decommission      | Save SaaS Upgrade Assistant deploy result CSVs, entity inventory spreadsheets, and configuration mapping documents before decommissioning Managed.                                                                             | Recommended |
| Validate all dashboard tiles show data               | Open every migrated dashboard. Tiles with "No data" indicate missing entity mappings or incorrect management zone IDs.                                                                                                         | Recommended |
| Set API token expiration dates                       | Set expiration on all tokens. Store in secrets management (HashiCorp Vault, AWS Secrets Manager). Rotate regularly.                                                                                                            | Recommended |



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